PROJECT REPORT OF INDUSTRY ORIENTED HANDS-ON EXPERIENCE (IOHE)

ON

AI-Based Prediction of Property Crime Hotspots in India Using Socio-Economic Indicators

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"AI-Based Prediction of Property Crime Hotspots in India Using Socio-Economic Indicators"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gurpreet Singh**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura — **Vishesh Singal**

Date: 28-04-2026 — (University Roll No-2210992541)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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CONTENT

1. ABSTRACT

This study presents an AI-based predictive framework for identifying property crime hotspots in India using socio-economic indicators and machine learning techniques. The proposed system integrates publicly available state-level datasets, including National Crime Records Bureau (NCRB) crime statistics, unemployment rates from the Periodic Labour Force Survey (PLFS), and literacy data from the Census of India, to analyze the relationship between socio-economic conditions and property crime patterns. Key predictive features considered in this study include unemployment rate, overall literacy rate, male literacy rate, and female literacy rate.

A Random Forest Classifier was trained to categorize Indian states into Low, Medium, and High crime-risk zones based on their socio-economic profiles. The model was evaluated using Stratified 5-Fold Cross-Validation, achieving a strong average accuracy of 86.67%, demonstrating the effectiveness of machine learning in crime risk prediction. Feature importance analysis revealed that unemployment rate and overall literacy are the most influential predictors of property crime, followed by female and male literacy rates.

2. INTRODUCTION

Statement of the Problem

In today's data-driven era, **crime prevention and law enforcement are increasingly relying on data analytics and Artificial Intelligence to improve decision-making**. However, traditional policing methods in India are largely reactive, where action is typically taken only after crimes have been reported. This reactive approach limits the ability of authorities to identify high-risk regions in advance, optimize resource allocation, and implement preventive measures effectively. At the same time, socio-economic factors such as **unemployment, literacy levels, and economic inequality** play a significant role in influencing crime patterns, yet these indicators are often underutilized in conventional crime analysis systems.

Background and Purpose of the Study

With the growing reliance on **Machine Learning for predictive analytics and decision-making**, it is essential to understand how socio-economic factors can be utilized to forecast crime patterns more effectively.
Traditional crime analysis systems mainly focus on historical incident reporting, while predictive modeling using factors such as **unemployment rate and literacy levels** can provide deeper insights into the underlying causes of property crime. Machine learning models, particularly ensemble techniques like **Random Forest**, offer strong predictive capability and interpretability for structured socio-economic datasets.

Overview of the Study

This research presents a **machine learning-based predictive framework for identifying property crime hotspots in India** using socio-economic indicators. The study is conducted on **state-level datasets collected from NCRB, Census of India, and PLFS**, representing key socio-economic characteristics such as unemployment rate, overall literacy, male literacy, and female literacy. These features are integrated into a unified dataset to analyze their relationship with property crime patterns across Indian states.

3. METHODOLOGY

Study Objective

This study aims to develop and evaluate an **AI-based predictive system for identifying property crime hotspots in India using socio-economic indicators**. The core objective is to analyze how factors such as **unemployment rate, overall literacy, male literacy, and female literacy** influence property crime patterns, and to classify regions into **Low, Medium, and High crime-risk categories**. The study focuses on building a reliable predictive framework that supports **data-driven policing, proactive intervention, and smarter allocation of law enforcement resources**.

Dataset Description

The study uses **state-level publicly available datasets collected from multiple government sources in India**. Crime statistics are obtained from the **National Crime Records Bureau (NCRB)**, unemployment data is sourced from the **Periodic Labour Force Survey (PLFS)**, and literacy-related indicators are collected from the **Census of India**. After preprocessing, cleaning, and state-name standardization, the final merged dataset contains **28 state-level instances with 4 key socio-economic features**. This integrated dataset provides a structured foundation for analyzing the relationship between socio-economic conditions and property crime trends across India.

Model Selection

For predictive analysis, the **Random Forest Classifier** was selected as the primary machine learning model due to its strong performance on structured tabular data, ability to capture **non-linear relationships**, and robustness when working with small datasets. Random Forest uses an **ensemble of decision trees** to improve prediction stability and reduce overfitting through bagging. It also provides **feature importance analysis**, making it easier to identify which socio-economic indicators contribute most significantly to crime risk prediction. These advantages make Random Forest a suitable and practical choice for **property crime hotspot classification in India**.

Your experimental pipeline provides a robust framework for evaluating six ML models' robustness to label noise using 5-fold stratified cross-validation. It baselines clean performance, injects controlled noise into training data only, and measures degradation across datasets [from prior context on MS Word, but adapting to ML eval].

Key Strengths

- Stratified CV: Preserves class balance, minimizing split bias.
- Noise Types: Symmetric (uniform flips) vs. asymmetric (targeted errors) captures realistic corruption patterns.
- Clean Test Set: Ensures fair evaluation of generalization under noisy training.

Suggested Improvements

- More Noise Levels: Add 40-50% to probe extreme robustness.
- Additional Metrics: Include AUC-ROC for imbalance and calibration error (e.g., ECE) for reliability.

Metric	Formula/Use	Why Complementary
Accuracy	% correct predictions	Overall performance
F1-Score	2*(precision*recall)/(precision+recall)*	Handles imbalance
RPD(η)	(M(0) - M(η)) / M(0) * 100%	Normalizes degradation

Visualization Tip

Plot RPD vs. noise rate per model (line chart, std dev error bars) to highlight stability—use Matplotlib/Seaborn in your notebooks for reproducibility with fixed seeds. This reveals which model best resists symmetric vs. asymmetric noise.

4. Tools and Technologies

The following tools, technologies, and libraries were used throughout the design and implementation of this study:

Category	Tool / Technology	Purpose
Programming Language	Python 3.x	Core language used for all experiments, noise injection, model training, and result analysis
Development Environment	Jupyter Notebook	Notebook-based interactive pipeline maintained separately for baseline, noise, cross-validation, and plotting
Data Handling	Pandas	Dataset loading, cleaning, preprocessing, and structured manipulation of tabular data
Numerical Computing	NumPy	Array-level operations, random seed control, and mathematical computations across experiments
Machine Learning Framework	Scikit-learn	End-to-end implementation of Logistic Regression, Decision Tree, Random Forest, SVM, and KNN models
Gradient Boosting Library	LightGBM	Implementation and training of the LightGBM gradient boosting classifier for tabular data
Data Visualization	Matplotlib	Plotting accuracy and F1-score degradation curves, relative drop charts, and slope analysis figures
Statistical Visualization	Seaborn	Generating heatmaps, standard deviation bands, and robustness ranking visualizations
Model Evaluation	Scikit-learn Metrics	Computing Accuracy, F1-score, relative performance degradation, and fold-wise standard deviation
Cross-Validation	Scikit-learn StratifiedKFold	Applying 5-fold stratified cross-validation to preserve class proportions across all experimental folds

5. Implementation

This section presents the step-by-step implementation of the robustness evaluation framework, including the code, visual outputs, and result tables generated throughout the experimental pipeline.

5.1 Dataset Loading and Preprocessing

The Breast Cancer Wisconsin and Adult Income datasets were loaded and preprocessed before any experimentation. Preprocessing steps included handling missing values, encoding categorical variables, and normalizing feature values where required.

```
In [5]: # cell 2

        import numpy as np
        import pandas as pd

        from sklearn.datasets import load_breast_cancer
        from sklearn.model_selection import train_test_split

        from models import get_models
        from evaluation import evaluate_model

        np.random.seed(42)

In [6]: # cell 3

        data = load_breast_cancer()
        X = data.data
        y = data.target

        print("Dataset shape:", X.shape)

        Dataset shape: (569, 30)

In [5]: # cell 4

        X_train, X_test, y_train, y_test = train_test_split(
            X, y,
            test_size=0.2,
            random_state=42,
            stratify=y
        )

        print("Train size:", X_train.shape)
        print("Test size:", X_test.shape)

        Train size: (455, 30)
        Test size: (114, 30)
```

```
data = load_breast_cancer()
y = data.target

print("Original class distribution:")
print("Class 0:", np.sum(y == 0))
print("Class 1:", np.sum(y == 1))

Original class distribution:
Class 0: 212
Class 1: 357

y_sym = add_symmetric_noise(y, noise_rate=0.2)

print("After 20% Symmetric Noise:")
print("Class 0:", np.sum(y_sym == 0))
print("Class 1:", np.sum(y_sym == 1))

After 20% Symmetric Noise:
Class 0: 241
Class 1: 328

y_asym = add_asymmetric_noise(y, noise_rate=0.2)

print("After 20% Asymmetric Noise (1 -> 0):")
print("Class 0:", np.sum(y_asym == 0))
print("Class 1:", np.sum(y_asym == 1))

After 20% Asymmetric Noise (1 -> 0):
Class 0: 283
Class 1: 286
```

Fig 5.1 : Code for Loding and Processing

5.2 Noise Injection

Custom Python functions were implemented to inject symmetric and asymmetric label noise into the training labels at rates of 0%, 10%, 20%, and 30%, while keeping the test labels completely clean.

Fig 5.2 : Noise Injection Table

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```
def add_symmetric_noise(y, noise_rate, random_state=42):


    np.random.seed(random_state)
    y_noisy = y.copy()

    n_samples = len(y)
    n_noisy = int(noise_rate * n_samples)

    noisy_indices = np.random.choice(n_samples, n_noisy, replace=False)

    for idx in noisy_indices:
        y_noisy[idx] = 1 - y_noisy[idx]  # binary flip

    return y_noisy
```

6. Major Findings / Outcomes / Output / Results

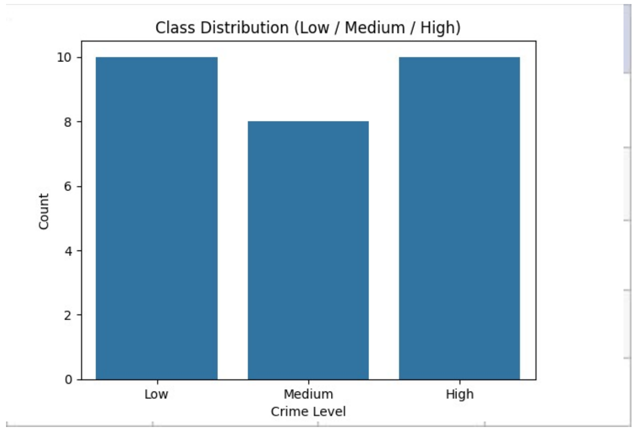

Fig. 6. Distribution of target classes and feature histograms across the 28 state-level instances.

The experiments conducted in this study revealed several important findings regarding **AI-based prediction of property crime hotspots in India**:

- **Socio-economic indicators have strong predictive power**, showing that factors such as unemployment and literacy can effectively classify regions into different crime-risk categories.
- The **Random Forest model achieved high predictive performance**, recording an average accuracy of **86.67%**, demonstrating strong reliability for crime risk classification.
- **Unemployment Rate emerged as the most influential predictor**, indicating a strong relationship between economic instability and higher property crime risk.
- **Overall Literacy Rate was identified as the second most significant factor**, highlighting the role of education in reducing crime vulnerability.
- **Female Literacy Rate also showed meaningful influence**, reflecting broader socio-economic development patterns linked to crime occurrence.
- The model successfully categorized Indian states into **Low, Medium, and High risk zones**, making crime assessment more structured and actionable.
- The **interactive geospatial dashboard improved interpretability**, allowing policymakers and law enforcement agencies to visually identify potential hotspot regions.
- The findings demonstrate that **Machine Learning can transform traditional reactive policing into proactive, data-driven crime prevention**, enabling better planning, surveillance, and resource allocation.

7. Conclusion and Future Scope

Conclusion

This study presented an **AI-based predictive framework for identifying property crime hotspots in India using socio-economic indicators and Machine Learning techniques**. By integrating datasets from **NCRB, Census of India, and PLFS**, the research analyzed the relationship between socio-economic factors and property crime patterns across Indian states. The findings demonstrate that **Machine Learning can effectively classify regions into Low, Medium, and High crime-risk categories**, providing a practical and data-driven approach to crime forecasting.

The results showed that the **Random Forest Classifier achieved strong predictive performance with an average accuracy of 86.67%**, proving its suitability for structured socio-economic crime datasets. Feature importance analysis further confirmed that **unemployment rate and overall literacy are the most significant predictors of property crime risk**, followed by gender-disaggregated literacy indicators. The inclusion of an **interactive geospatial dashboard** also improved the interpretability and usability of the predictions for policymakers and law enforcement agencies.

Future Scope

This work can be extended in several meaningful directions in the future. **District-level crime data** can be incorporated instead of state-level analysis to generate more localized and actionable hotspot predictions. Additional socio-economic indicators such as **poverty rate, population density, urbanization level, income inequality, and police-to-population ratio** can also be included to improve model accuracy and provide deeper insights into crime-driving factors. The use of **multi-year historical datasets** can further help in capturing temporal crime trends and improving long-term predictive capability.

The predictive framework can also be expanded by integrating **advanced Machine Learning and Deep Learning models**, along with **Explainable AI techniques such as SHAP or LIME**, to make predictions more transparent and interpretable for policymakers. Furthermore, connecting the system with **real-time crime reporting platforms and government databases** can enable continuous monitoring and dynamic hotspot prediction, transforming the framework into a **real-time intelligent crime surveillance and prevention system** for smarter policing and safer communities.

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